Supplement: parameter-sensitivity methods

This supplement documents how the sensitivity grid of sec. 17.13 is generated and — importantly for a reader trying to reconcile numbers across studies —

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Supplement... (part 2)

where its two sweeps use *different* seeding and trial budgets. It answers two questions the main section leaves implicit: exactly which seed drives each cell, so every cell is deterministically reproducible in isolation;

Supplement... (part 3)

and why the cross-study summary's sensitivity row is not directly comparable, at matched trial counts, to the standalone heatmap.

Experimental protocol for grid sensitivity

The sensitivity sweep uses `run_belief_sharing_sensitivity` and `run_hierarchical_sensitivity` from the `fedference.experiments` module.

Experimental protocol for… (part 2)

Each function accepts a tuple of acuity values and a tuple of colony sizes. In the **belief-sharing** sweep every (acuity, colony-size) cell averages n_{trials} independent trials, each seeded via a deterministic formula:

Experimental protocol for... (part 3)

$$\mathsf{seed}_{\mathsf{cell}} = \mathsf{seed}_{\mathsf{base}} + i \cdot 10^5 + j \cdot 10^3 + t \quad (40)$$

Experimental protocol for... (part 4)

where i indexes acuity, j indexes colony size, and t indexes the trial within a cell. The deterministic rule eq. 40 makes every grid cell and trial reproducible in isolation from the base seed. For the belief-sharing sweep, no two cells share a trial seed,

Experimental protocol for… (part 5)

so the design introduces no seed reuse between cells.

Re-running with the same `seed_base` is bit-identical. Changing `seed_base` produces an independent replicate for robustness checking.

Experimental protocol for... (part 6)

The **hierarchical** sweep uses a simpler protocol: every cell calls `run_hierarchical_world` once with the same base seed (its internal trials are seeded by that run), so hierarchical cells share the base seed rather than the per-cell formula above.

Grid parameters for acuity and colony size

Parameter	Values
Sensor acuity κ	$\{0.40, 0.55, 0.70, 0.85, 0.95\}$
Colony size n	$\{2, 4, 6, 8, 10\}$

Grid parameters for acuity... (part 2)

The $5 \times 5 = 25$ cells per system are run with `seed_base = 0` by default; `generate_sensitivity_heatmap` accepts a `seed` argument to override this.

Belief-sharing condition in the sensitivity grid

Each trial follows four labeled operations. **Draw:** sample one true state and one noisy observation per agent using the `run_belief_sharing` protocol.

Belief-sharing condition in... (part 2)

Compare: run one belief-sharing round with `communicate=True` and the matched isolated condition with `communicate=False`. **Record:** retain `mean_accuracy` for both conditions.

Belief-sharing condition in… (part 3)

Reduce: define the cell value as the mean communicating-minus-isolated gap across `n_trials`.

Hierarchical POMDP condition in the sensitivity grid

Each cell in the hierarchical sweep calls `run_hierarchical_world` once with the cell's acuity and colony size, passing the constant base seed (not the per-cell formula, which applies only to the belief-sharing sweep).

Hierarchical POMDP... (part 2)

The returned `location_accuracy_gap` (hierarchical minus flat) becomes the cell value.

Figure rendering for sensitivity summaries

`generate_sensitivity_heatmap` assembles the two grids into vertically stacked matplotlib `imshow` panels with a color-vision-deficiency-safer blue–neutral–brown signed palette and symmetric bounds at $\pm \max(|\text{gap}|)$,

Figure rendering for… (part 2)

per-cell numeric annotations, and a per-panel colorbar labeled “Accuracy difference”. The figure is written as `sensitivity_heatmap.png` under `output/figures/`. Negative gaps map toward blue, positive gaps toward brown,

Figure rendering for... (part 3)

and zero to the neutral midpoint; exact signed cell labels duplicate that encoding. Hatching marks only the declared $|\mathrm{gap}| \leq 0.05$ display band. It is not an unreliability flag, confidence interval, significance test,

or proof of zero effect.

Cross-study summary construction

`generate_cross_study_summary` performs a separate harmonized 128-seed ($n_{\text{seeds}} = 128$) rerun over Studies 1–9 and reports the mean with a 95% seed-level percentile-bootstrap interval for each headline signed contrast or estimand. These intervals belong to that rerun,

Cross-study summary... (part 2)

including for rows whose primary figure is a deterministic single-posterior or fixed-configuration diagnostic; they are not intervals on the primary deterministic display. The definitions are:

Cross-study summary... (part 3)

The robustness row uses 40 matched trials per seed and rate; the trial-level observations are reduced within seed before the cross-study summary is formed. Within each seed, the row then takes the display-selected maximum non-reference server-preset contrast at the declared worst rate.

Cross-study summary... (part 4)

This preserves the seed as the independent Monte Carlo unit but makes Study 4 a within-run display-selected summary, not a preselected method or inferential winner.

Cross-study summary… (part 5)

The Study 8 row below uses 3 trials per cell — smaller than the full-resolution 20-trial `Trials per cell` grid documented above for the standalone sensitivity heatmap figure, a deliberate runtime budget for the per-seed cross-study loop rather than an oversight —

Cross-study summary... (part 6)

so the two are not directly comparable at matched trial counts.

Cross-study summary… (part 7)

Study	Metric
1 — Belief sharing	Accuracy contrast: communicating - isolated
2 — Language acquisition	KL reduction: initial - final

Cross-study summary… (part 8)

Bootstrap CIs use 5000 resamples. Their default `n_boot` is defined by `bootstrap_ci` in the `fedference.statistics` module.